Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Date of Assignment :12-08-2026	Date/Time of Submission :12-08-2026

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

Code:

```
# Bubble sort in Python

def bubbleSort(array):

  # loop to access each array element
  for i in range(len(array)):

    # loop to compare array elements
    for j in range(0, len(array) - i - 1):

      # compare two adjacent elements
      # change > to < to sort in descending order
      if array[j] > array[j + 1]:

        # swapping elements if elements
        # are not in the intended order
        temp = array[j]
        array[j] = array[j+1]
        array[j+1] = temp


data = [-2, 45, 0, 11, -9]

bubbleSort(data)

print('Sorted Array in Ascending Order:')
print(data)

print("NirajRane_S032")
```

Output:

```
---------------------------------------------------- RESTART: C:/
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
NirajRane_S032
```

b. Insertion sort

Code:

```python
# Insertion sort in Python


def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key


data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)

print("NirajRane_S032")
```

Output:

```
----------------------------------------
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
NirajRane_S032
```

c. Selection sort

Code:

```
# Selection sort in Python


def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])


data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)

print("NirajRane_S032")
```

Output:

```
==========================================
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
NirajRane_S032
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.

   Code:

```
# Bubble sort in Python

def bubbleSort(array):

  # loop to access each array element
  for i in range(len(array)):

    # loop to compare array elements
    for j in range(0, len(array) - i - 1):

      # compare two adjacent elements
      # change > to < to sort in descending order
      if array[j] < array[j + 1]:

        # swapping elements if elements
        # are not in the intended order
        temp = array[j]
        array[j] = array[j+1]
        array[j+1] = temp


data = [-2, 45, 0, 11, -9]

bubbleSort(data)

print('Sorted Array in Descending Order:')
print(data)

print("NirajRane_S032")
```

   Output:

```
========================================
Sorted Array in Descending Order:
[45, 11, 0, -2, -9]
NirajRane_S032
>
```

2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.

Code:

```python
# Insertion sort in Python


def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key > array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key


data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Descending Order:')
print(data)

print("NirajRane_S032")
```

Output:

```
=============================================
Sorted Array in Descending Order:
[9, 5, 4, 3, 1]
NirajRane_S032
>>>
```

3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.

Code:

File Edit Format Run Options Window Help

```
# Selection sort in Python


def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] > array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])


data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Descending Order:')
print(data)

print("NirajRane_S032")
```

Output:

```
========================================
Sorted Array in Descending Order:
[45, 11, 0, -2, -9]
NirajRane_S032
```